

# MERYEM OUAFI

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## ABOUT ME

I'm currently a second-year master's student in Data Science and Smart Systems. I'm now seeking an end-of-studies internship (PFE) to apply my skills in machine learning, data analysis, and big data technologies to impactful projects, learn from real-world experiences, and contribute to a team focused on data-driven decision-making.

## SKILLS

### Programming & Development

Python (Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, Keras, Flask, Django), SQL, Prolog, Java

### SQL

MySQL, PostgreSQL, SQLite, Microsoft SQL Server

### NoSQL

MongoDB, Data manipulation, Replication, Sharding for large-scale data distribution

### Big Data

Hadoop (HDFS, MapReduce), YARN, Hive, Pig, Spark, Sqoop, Flume, Kafka, HBase, MapReduce development, Pig Latin.

### Data Analysis & Statistical Modeling

Linear regression (simple and multiple), KNN (multiclass classification), Logistic regression (binary classification), Unsupervised clustering.

### Data Mining

CRISP-DM, Problem definition, Data collection & understanding, Data preparation (cleaning, missing values handling), Algorithm selection, Model evaluation, Results interpretation, Analytical solution deployment.

### Artificial Intelligence

Propositional and predicate logic, Knowledge representation (Prolog), Graph-based problem solving, Planning, Expert systems, Inference engines, Constraint Satisfaction Problems (CSPs) modeling and solving

### Natural Language Processing

Text preprocessing, Tokenization, Word embeddings, Summarization, Translation, Question answering, Generation, Speech recognition.

## ACADEMIC PROJECTS

### Improving Emergency Services Based on 911 Call Data

- Enhanced emergency services using the CRISP-DM model.
- Conducted data collection, exploration, preparation, model building, evaluation, and deployment.
- Optimized Random Forest, Decision Tree, and Gradient Boosting models for better prediction accuracy.
- Deployed the solution in a real-world application for resource optimization.
- **Technologies:** Python, Pandas, Scikit-learn, Matplotlib, Jupyter Notebook.

**Link** https://github.com/mereymouafi/Improving-Emergency-Services-Based-on-911-Call-Data

### Mobile Price Range Prediction

- Predicted mobile phone prices through multiple linear regression, analyzing relationships between specifications and price.
- **Technologies:** Python, NumPy, Pandas, Scikit-learn, Matplotlib.

**Link** https://github.com/mereymouafi/Mobile-Price-Range-Prediction

### Log Analysis with Apache Pig

- Performed large-scale data analysis using Hadoop and HDFS.
- Created Apache Pig scripts for log analysis, including segmenting requests by browser and counting IP requests.
- **Technologies:** Apache Hadoop, Apache Pig, HDFS, Bash.

**Link** https://github.com/mereymouafi/Log-Analysis-with-Apache-Pig

### Implementation of Apache Giraph

- Analyzed Apache Giraph architecture, focusing on master/worker nodes and ZooKeeper integration.
- Configured Giraph for distributed graph processing, including supersteps and replication.
- **Technologies:** Apache Giraph, ZooKeeper, Java

## EDUCATION

2023 – 2025
### M.S in Data Science and Smart Systems

Mohammed First University - Multidisciplinary Faculty of Nador

2019 – 2022
### B.S in Mathematical and Computer Sciences

Mohammed First University - Faculty of Sciences, Oujda

2019 – 2021
### Diploma of General University Studies in Mathematical and Computer Sciences

Mohammed First University - Faculty of Sciences, Oujda

2018 – 2019
### Scientific Baccalaureate

High School Princesse Lalla Asmae

## EXPERIENCE

02/2023 – 05/2023
### Full-Time Intern - Ministry of Culture

- Developed an application for the digitization of records to streamline document management.
- Performed data entry and verification to ensure accuracy and consistency.
- Collaborated with the development team to enhance the quality and efficiency of the project.
- Utilized analytical and organizational skills to support project success.

## SOFT SKILLS

**Critical Thinking | Problem-Solving | Attention to Detail Teamwork | Adaptability | Conflict Resolution**